

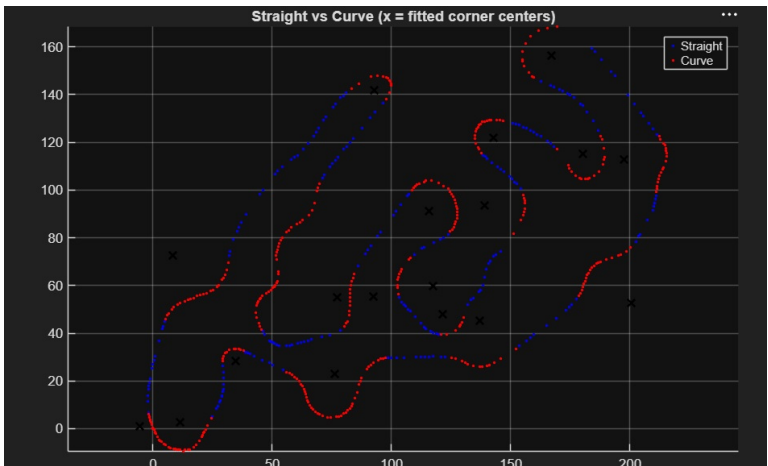
Lap Time Simulation: Methodology

Overview

The lap time simulation splits a lap into two physically different problems: cornering, where speed is limited by available grip, and straights, where speed is limited by the powertrain and braking system. Each is solved with a different model, then stitched together corner-to-straight-to-corner to build a full lap.

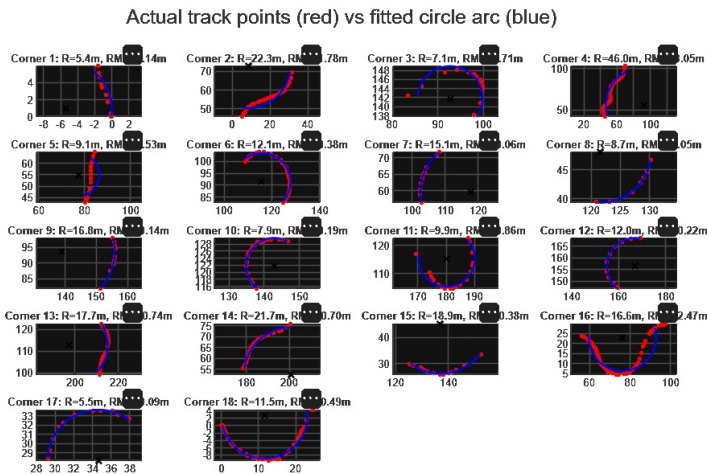
1. Track Reconstruction

Before any speed can be solved, the track itself has to be turned into geometry. Raw track points are classified as either straight or curved sections, then each curved section is fit to a circular arc to recover its radius and included angle.



Track segmentation

Corner centers (marked with an X) come out of this fit directly. This geometry, radius and angle per corner, is what feeds the cornering solve below.



Corner arc fits

Each panel compares the actual logged track points (red) against the fitted circular arc (blue), with fit radius and RMSE reported per corner. Some RMSE values are partially obscured by a UI overlay in the exported plot and weren't recoverable from this image; worth re-exporting cleanly if this goes into a final report.

2. Cornering Speed

For each corner, maximum speed is solved from a force balance rather than assumed. At a candidate speed, the model computes:

- Aerodynamic downforce at that speed, split evenly across all four wheels
- Lateral load transfer front and rear, from cornering acceleration, CG height, and track width
- Resulting vertical load at each wheel (static load, aero, and load transfer combined)
- Available lateral grip at each wheel from a load-sensitive friction model, where the friction coefficient itself decreases as vertical load increases, then derated by a fixed factor
- Total available lateral force, summed across all four wheels, versus the lateral force required to hold that speed through that corner's radius

The correct cornering speed is the one where required and available lateral force match. This is solved per corner with a root-finder rather than assumed from a fixed friction circle, so it accounts for how downforce and load transfer change with speed.

Corner time then follows directly from arc length and solved speed.

3. Straight-Line Simulation

Straights are not solved with the same quasi-steady-state approach. Entry and exit speed for each straight are taken directly from the corner solve on either side (a corner's solved speed becomes the entry speed for the straight leaving it, and the exit speed for the straight entering it), and the straight itself is simulated as a full transient: the car accelerates from its entry speed, then must brake down to its exit speed before the next corner.

The unknown is where along the straight the car switches from accelerating to braking. This transition point is found iteratively: a full vehicle acceleration model is run forward from the entry speed, and a separate braking model is integrated backward from the exit speed (as an ODE that includes aerodynamic drag, not a constant-deceleration assumption). The transition point is adjusted until the speed reached by the accelerating phase matches the speed required by the braking phase at the same point on the straight.

Total time on the straight is the sum of both phases.

4. Gear Ratio and Total Lap Time

The straight-line simulation is run across a set of candidate gear ratios, and the ratio giving the lowest total straight-line time is selected. Total lap time is the sum of cornering time (summed across all corners) and optimized straight-line time (summed across all straights).